A Structured Approach for Unsupervised Automatic Short Answer Grading

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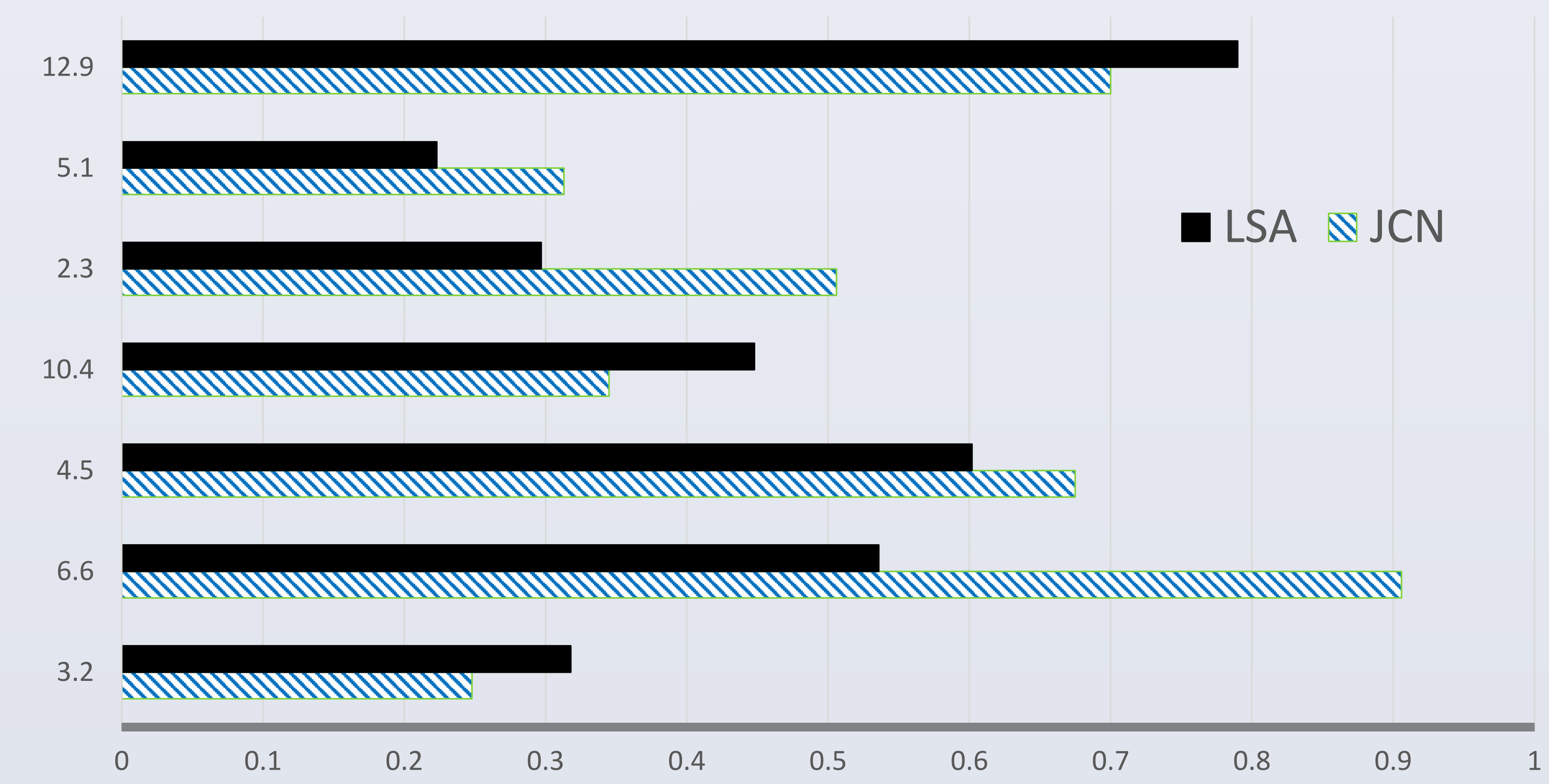
Problem Definition

- Systems designed to automatically assess short answers in natural language having a length of a few words to a few sentences

Question	How are overloaded functions differentiated by the compiler?
Model Answer	Based on the function signature. When an overloaded function is called, the compiler will find the function whose signature is closest to the given function call.
Student Answer 1	It looks at the number, types, and order of arguments in the function call.
Student Answer 2	By the number, and the types and order of the parameters.

Motivation

Variation of Pearson r correlation scores across question types

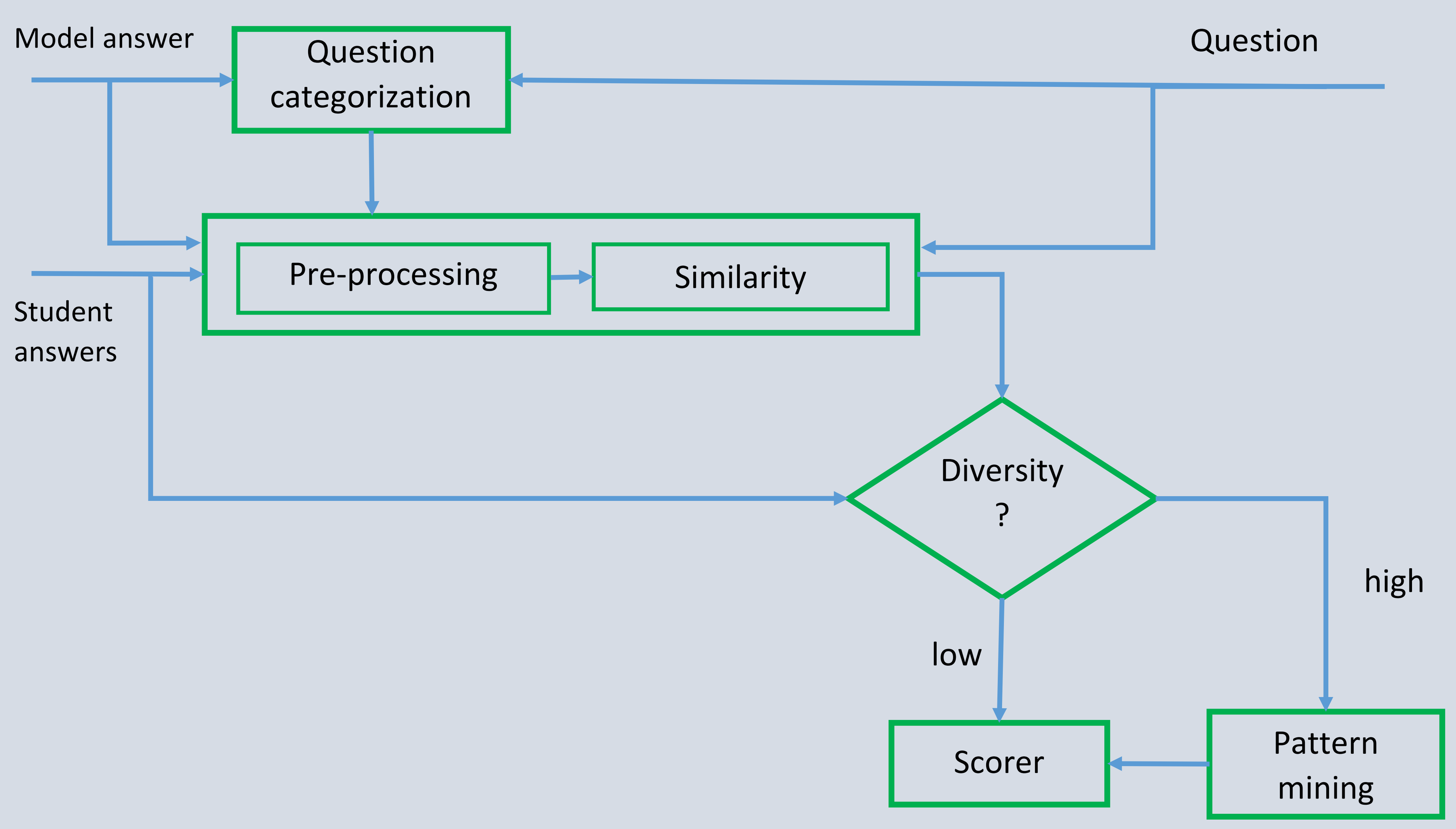


We have adopted Graesser's question taxonomy and each question on y-axis in above graph belongs to the following categories, Disjunctive, Concept Completion, Feature Specification, Quantification, Definition, Comparison, Interpretation and Procedural. Observing the variations in Pearson's r score across question categories we hypothesize that using question categories would improve the performance of ASAG techniques.

Approach

- Providing recommendations based on the knowledge of question category and diversity in student answers
- Adapted Graesser's question taxonomy
- Developed techniques natural to question categories along with suitable preprocessing and similarity techniques (lexical, knowledge-based and vector-spaced)

Showing a approximate pipeline of steps involved in our approach.



Note: The above diagram shows a basic pipeline (for a general case) though for some categories the exact algorithm could differ slightly.

Techniques

We have designed algorithms for each category but highlighting some of the key algorithms,

Pattern Mining:

- A variant of sequential pattern mining to exploit the high degrees of lexical overlap among student answers
- Works as a feedback based algorithm for questions with hi-diversity

Example:

1. What is a linked list? (Hi-diversity)
2. What is a leaf?

Comparison questions:

For these questions, we have extracted the entities from questions and used them to identify their attributes in reference and student answers. These entity-attribute pairs have been used for evaluation. For example,

Question: What is the difference between a data member and a local variable inside a member function?

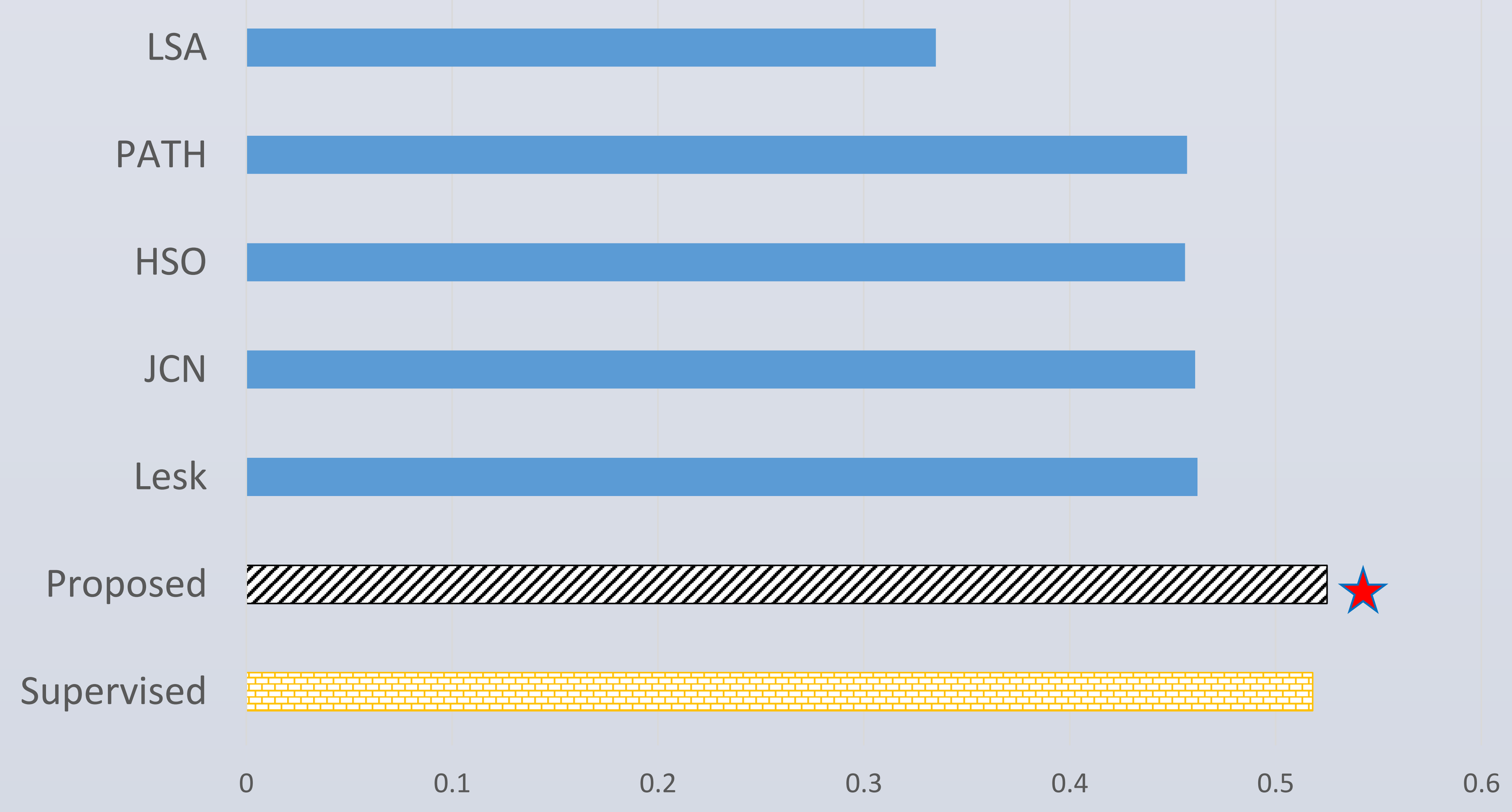
Reference Answer: Data members can be accessed from any member functions inside the class definition. Local variables can only be accessed inside the member function that defines them.

A snapshot of the parse tree obtained by using Stanford Parser shows the entities and attributes of the first sentence in the above reference answer.

```
(ROOT
  (S
    (NP (NNP Data) (NNS members))
    (VP (MD can)
      (VP (VB be)
        (VP (VBN accessed)
          (PP (IN from)
            (NP
              (NP (DT any) (NN member) (NNS functions))
              (PP (IN inside)
                (NP (DT the) (NN class) (NN definition)))))))
    (. .)))
```

Results

An improvement of 13% over earlier state-of-art unsupervised techniques on CSD dataset (Mohler et. al. 2011). This initial work has been submitted as a short paper to EMNLP 2016 and we are currently working on new techniques for better results.



References

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